

Agricultural Technology

Pre-National Diploma · Federal Polytechnic Wannune, Benue State, Nigeria · 2023

Approximately 60 students · Lecturer: Shagbaor Hycent Amool

Aim

To enable students to understand the sources of power available on a farm, how that power is generated, and how it is applied through machinery to the work of growing, harvesting, processing and storing crops.

Scope and basis

The course was taught at pre-diploma level and adapted from the National Board for Technical Education (NBTE) specification for *Farm Power and Mechanization* (AGT 223), National Diploma in Agricultural Technology. Topics follow that specification; depth and pace were adjusted for students entering the diploma programme.

Weekly schedule

Week	Topic	Covered
1	Sources of power on the farm	Human, animal, mechanical, wind, water, electrical and solar power; comparing energy sources on efficiency and cost.
2	Farm engines	Two-stroke and four-stroke engines; the difference between compression-ignition (diesel) and spark-ignition (petrol) operation.
3–4	Tractors	Tractor types and makes; steering, engine, transmission, final drive and implement control; matching power rating to the job; safe operation and routine maintenance.
5–6	Tillage machinery	Draft implements — subsoilers, mouldboard, disc, chisel and ridging ploughs, harrows; ground-driven and power-driven cultivators.
7–8	Planting and transplanting	Planters, seed drills and transplanters; calibrating a seed drill for a given crop.
9	Manure and fertiliser application	Machines for applying organic manures and artificial fertilisers; construction and working principles.
10	Crop protection equipment	Hand sprayers, boom sprayers and crop dusters; calibration and maintenance.
11–13	Harvesting machinery	Mowers and forage harvesters; pick-up balers; combine harvesters and their threshing components.
14–15	Processing and storage	Storage of grains, cereals, tubers, fruits and vegetables; primary processing equipment for local crops.

Teaching approach

Theory and practical work were taught together rather than separately, following NBTE guidance that practical exercises in professional courses should not be taught in isolation from theory. Classroom sessions were paired with workshop and field demonstration — engine testing, tractor identification and operation, implement handling and seed-drill calibration.

Assessment

Theory 50% · Practical 50%, in line with the NBTE evaluation scheme for the National Diploma in Agricultural Technology.

Prepared by Shagbaor Hycent Amool, Lecturer I, Federal Polytechnic Wannune (2022–2026, currently on study leave). Course content adapted from the National Board for Technical Education, *Curriculum and Course Specifications for the National Diploma in Agricultural Technology*, Kaduna, Nigeria. shagbaor.com